

Topic 2: Data, Information and Information Systems:

Data and Information

Data are raw facts, or unprocessed information presented in any form including text, numbers, images, video clips, and audio files.

Examples of data can be an date of birth, employee's name, number of hours worked in a week, inventory part numbers or sales orders.

Information is processed data, or data that has been given meaning or value. Information gives out knowledge to users with access to that particular information.

Management in an organisation can make informed decisions based on information, and not data.

Data acts as an input, while information is the output. Once processed, wrong data will give out wrong information, and vice versa.

Consider the following **scenario**:

You have been temporarily hired by the government of Malawi to participate in a survey which aims at determining the number of people affected by hunger due to heavy rains in the previous year. You are then given hard copy forms containing questions which you will use to ask your participants. You will need to record the responses from your participants on the same hard copy forms. Once completed, you are requested to enter the collected responses into a computer system for processing.

At the end of the whole exercise, you are asked to print a report from your computer which will be presented your your supervisor.

Application:

In the above scenario, the responses collected from the participants are data, and are fed as input into the computer system. The report printed after entering and processing the responses into the computer system is information, produced as output.

For better understanding, below is an illustration of the distinction between data and information.

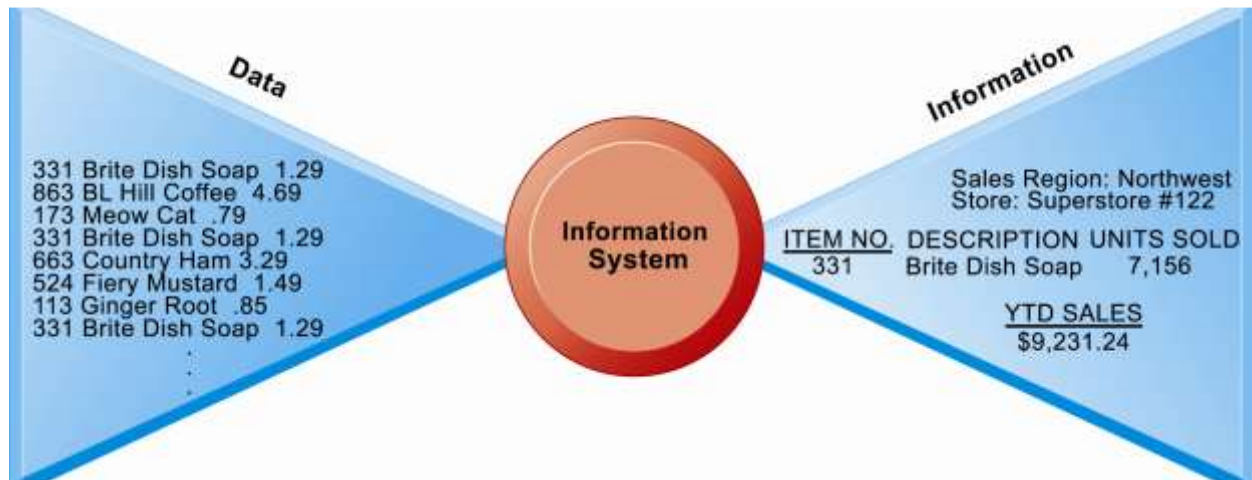


Figure 2.2: Data Vs. Information

And the pyramid below presents a data processing model:

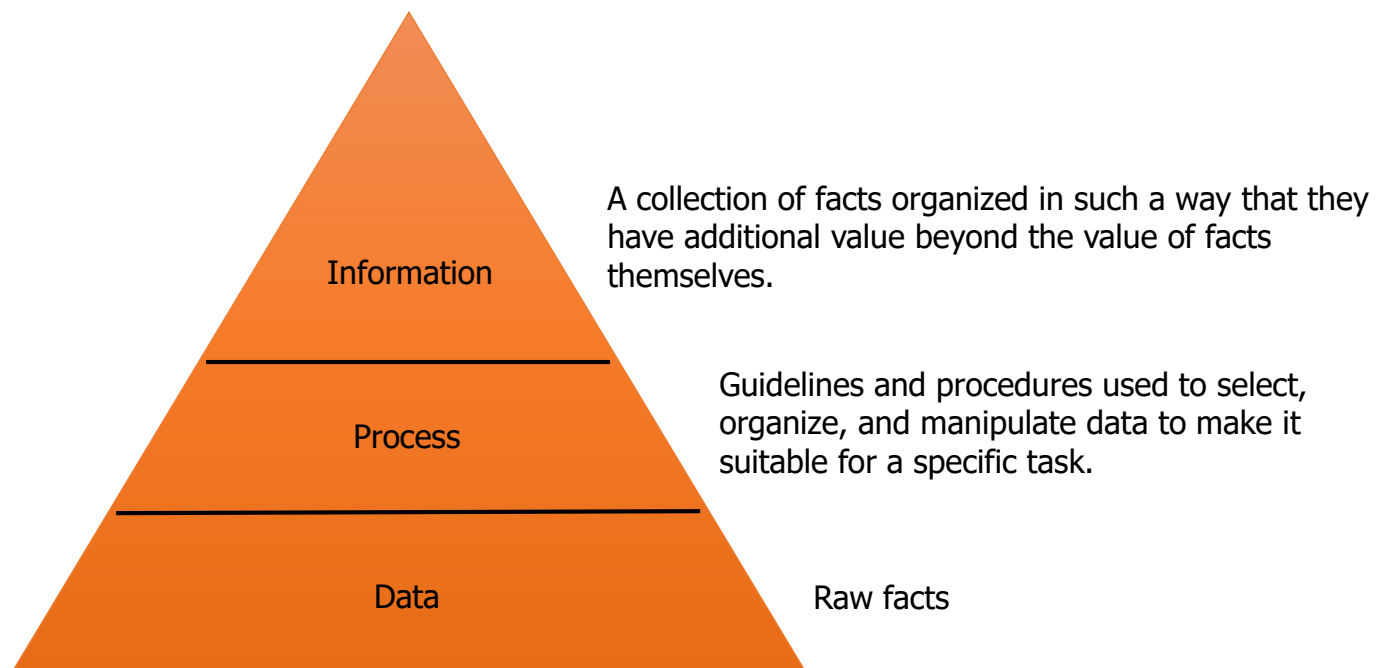


Figure 1: Data Processing Model

Data Transformation Process in Detail

Data transformation is the process of changing the format, structure, or values of data. For data analytics projects, data may be transformed at different stages.

Data is transformed to make it better-organized. Transformed data may be easier for both humans and computers to use.

Properly formatted and validated data improves data quality and protects applications from potential issues such as null values, unexpected duplicates, incorrect indexing, and incompatible formats.

Data transformation facilitates compatibility between applications, systems, and types of data. Data used for multiple purposes may need to be transformed through different ways.

Stages of Data Transformation

Transformation of data into information passes through different stages. They include the following:

- i. **Collection:** Involves collecting data from one or more sources.
- ii. **Preparation:** The collected data is organised and arranged ready for input.
- iii. **Input:** Data is entered into a computer system
- iv. **Processing:** Manipulating the data to produce the necessary output.
- v. **Output:** This is information, the final product after processing the input
- vi. **Storage:** Information is stored on storage devices in readiness for use.
- vii. **Utilisation:** Users make use of the stored information.

Characteristics of Information

- i. **Relevance:** Information should be communicated to the right person. It means a user should be make proper and informed decisions after acquiring knowledge from the available information.
- ii. **Completeness:** Information should be complete, meaning facts and figures should not be missing or concealed. Telling the truth but not wholly is of no use.

- iii. **Accuracy:** Information should be fair and free from bias. It should not have any arithmetical and grammatical errors.
- iv. **Timeliness:** Information should be communicated in time so that the receiver of the information has enough time to decide appropriate actions based on the information received.
- v. **Economical:** Information should be analysed for its benefits against the cost of obtaining it. In business context, it is not worthwhile to spend money on information that even cannot recover its costs leading to loss each time that information is obtained.

Information Technology (IT)

Information technology is defined as the set of tools used to receive, process, store, retrieve, and transmit information in an electronic form. Consists of all the hardware and software that firms need to use in order to achieve their business objectives.

Components of Information Technology

Hardware: This is the physical equipment used for input, processing, and output activities in an information system. It consists of the following: computers of various sizes and shapes (including mobile handheld devices); various input, output, and storage devices; and telecommunications devices that link computers together.

Software: Software are the detailed, preprogrammed instructions that control and coordinate the computer hardware components in an information system.

Databases: This is an organized collection of structured information, or data, typically stored electronically in a computer system.

Networking or telecommunications technology: This includes both physical devices and software, and connects various pieces of hardware. It is used to transmit data from one point to another. Computers and communications equipment can be connected in networks for sharing voice, data, images, sound, and video. A network links two or more computers to share data or resources, such as a printer.

Information Systems (IS)

An information system is a set of interrelated components that collect, manipulate, store data and disseminate information in an organisation.

It can also be defined as a set of interrelated components working together to collect, retrieve, process, store, and distribute information for the purpose of facilitating planning, control, coordination, analysis, and decision making in business organizations.

You must note that information technology is used to build an information system, meaning the set of tools in IT are necessary for the development of an IS.

Components of Information System

An information system is essentially made up of five components namely:

- i. Hardware
- ii. Software
- iii. Database
- iv. Network (Telecommunications)
- v. People (Human Resources)

These five components integrate to perform input, process, output, feedback and control.

Hardware consists of input and output devices, cpu, storage devices, and other components.

Software are detailed, preprogrammed instructions that control and coordinate the computer hardware components in an information system.

A **Database** is a collection of data organized to serve many applications efficiently by centralising the data and controlling redundant data.

Networking consists of communication media and network devices.

People consist of users who use the system. They include all company employees that have permission to access the information system, regardless of their positions or status.

Main Activities in an Information System

Refer to the diagram for the main activities that take place in an information system.

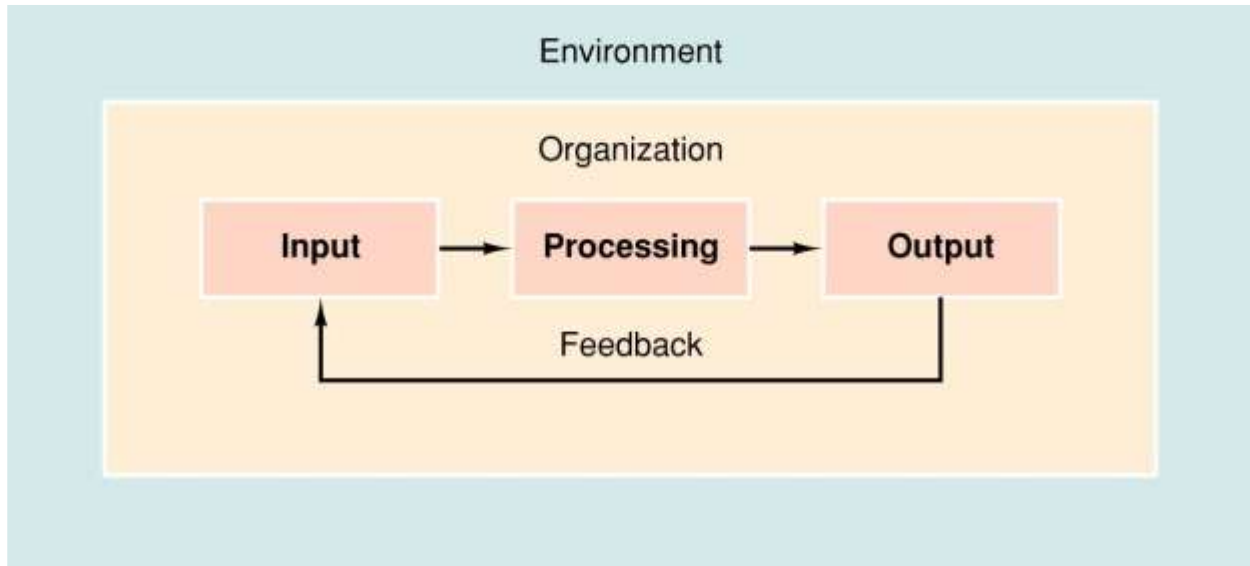


Figure 2: Main information system activities

As can be seen from the diagram above, the main activities in an information system are input, processing, and output. Some systems also have feedback, while others do not. These activities coordinate to produce the information that organizations need to make decisions, control operations, analyze problems, and create new products or services.

Input captures or collects raw data from within the organization or from its external environment. **Processing** converts this raw input into a meaningful form. **Output** transfers the processed information to the people who will use it or to the activities for which it will be used. And lastly, **Feedback** is output that is returned to appropriate members of the organization to help them evaluate or correct the input stage.

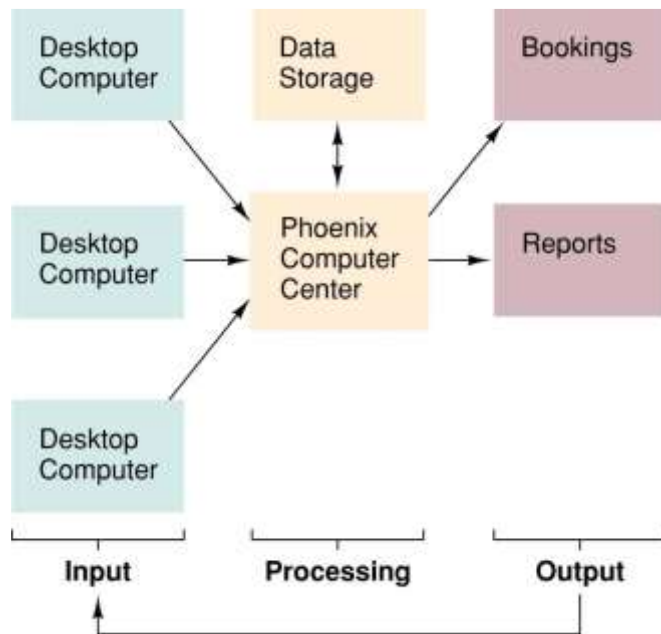


Figure 3: Example of an Information System

Information Technology Vs. Information System

For a better understanding on the relationship between information technology and information systems, refer to the diagram below.

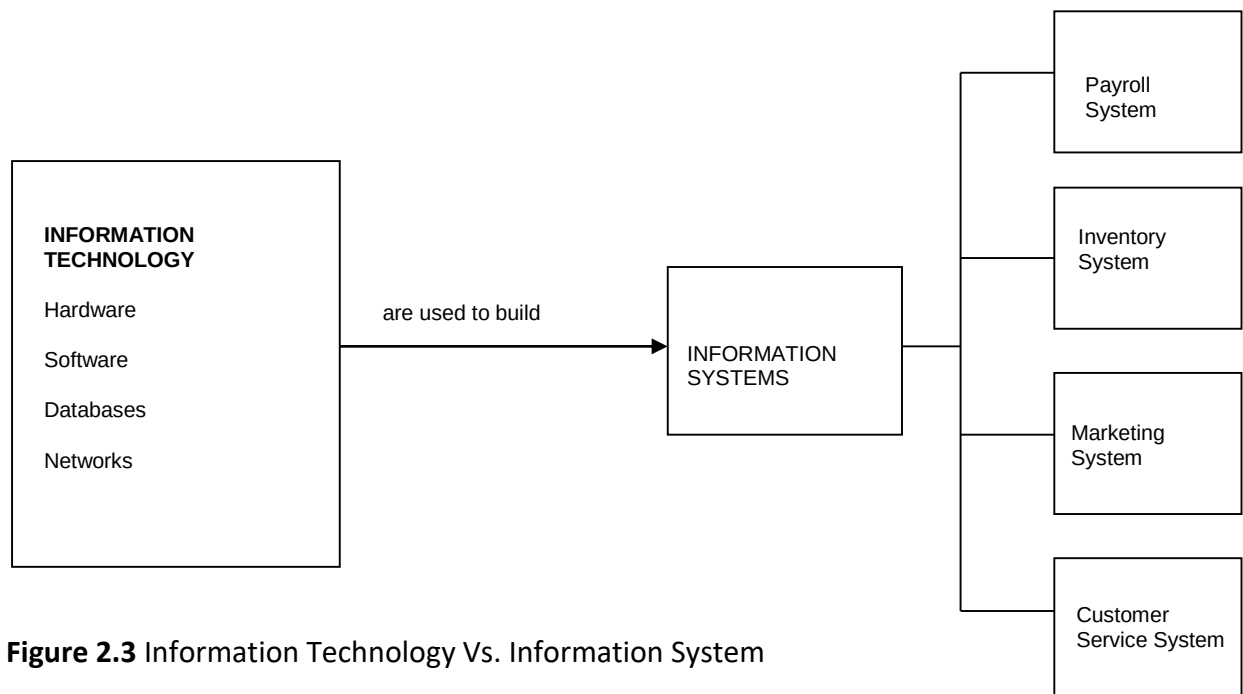


Figure 2.3 Information Technology Vs. Information System

IT Infrastructure for Information Systems

Information Technology infrastructure provides the foundation, or platform, on which the firm can build its specific information systems. Each organization must carefully design and manage its IT infrastructure so that it has the set of technology services it needs for the work it wants to accomplish with information systems. If not carefully planned, firms may end up acquiring information systems which will not add any value to their business processes.

Information Technology Infrastructure are the shared technology resources that provide a platform for the firm's specific information system applications. It can also be defined as a set of physical devices and software required to operate the entire enterprise. An IT infrastructure consists of a set of firm-wide services which are budgeted by management, comprising both human and technical capabilities.

IT Infrastructure Services

IT Infrastructure services include the following:

- Computing platforms providing computing services
- Telecommunications services
- Data management services
- Application software services
- Physical facilities management services
- IT management services
- IT education services
- IT research and development services

Components of IT Infrastructure

- i. Computer hardware platforms
- ii. Operating system platforms
- iii. Enterprise software applications
- iv. Data management and storage

- v. Networking/telecommunications platforms
- vi. Internet platforms
- vii. Consulting system integration services

Current Trends in Computer Hardware Platforms

Contemporary Hardware Platform Trends

- i. The Mobile Digital Platform
- ii. Consumerization of IT and BYOD
- iii. Grid Computing
- iv. Virtualization
- v. Cloud Computing
- vi. Green Computing
- vii. High-Performance and Power-Saving Processors

NOTE: Check for definitions of the above platforms in the prescribed text book.

Contemporary Software Platform Trends

- i. Linux and open source software
- ii. Software: Java, HTML, and HTML5
- iii. Web services and service-oriented architecture
- iv. Software outsourcing and cloud services

IT Management Issues/ Challenges

- i. Dealing with Platform and Infrastructure Change
- ii. Management and Governance
- iii. Making wise Infrastructure Investments

Overcoming Challenges

Total Cost of Ownership of Technology Assets

The actual cost of owning technology resources includes the original cost of acquiring and installing hardware and software, as well as ongoing administration costs for hardware and software upgrades, maintenance, technical support, training, and even utility and real estate costs for running and housing the technology.

The **total cost of ownership (TCO)** model can be used to analyze these direct and indirect costs to help firms determine the actual cost of specific technology implementations. The table describes the most important TCO components to consider in a TCO analysis. When all these cost components are considered, the TCO for a system might run up to three times the original purchase price of the equipment.

Competitive Forces Model for IT Infrastructure Investment

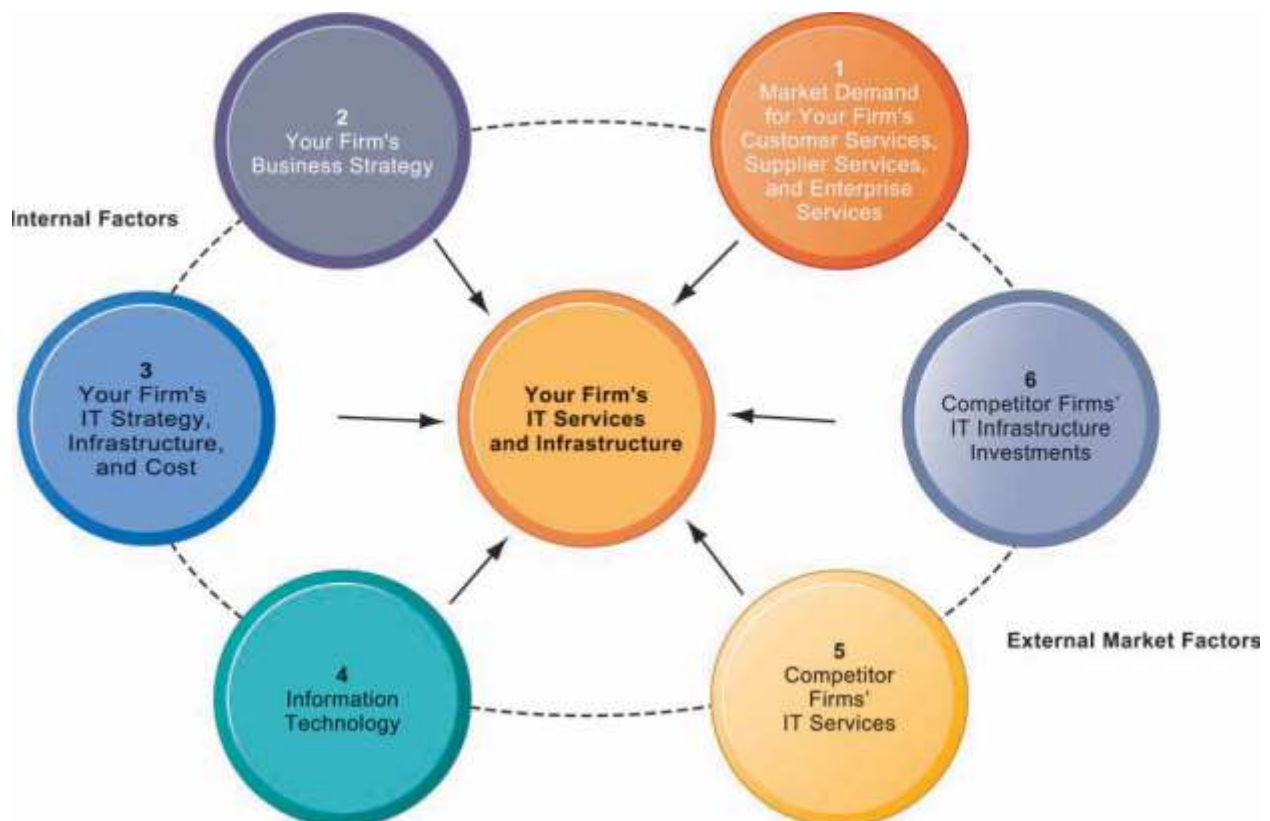


Figure 4: Competitive Forces Model for IT Infrastructure

Market demand for your firm's services: A firm should be able to make an inventory of the services it offers or provides to customers, suppliers, and employees. The firm may survey each group, or hold focus groups to find out if the services it currently offers are meeting the needs of each group.

A very good example would be to check whether customers complaining of slow responses to their queries about price and availability. A number of questions can be asked including: Are employees complaining about the difficulty of finding the right information for their jobs? Are suppliers complaining about the difficulties of discovering your production requirements? Solutions to these questions would help to know the demand for your firm's services or products.

Business strategy: An organisation is supposed to analyse its long time business strategies (like 5 years) and try to assess what new products, services and capabilities will be required to achieve its strategic goals.

IT strategy, infrastructure, and cost. A business firm must examine its information technology plans for the next five years and assess its alignment with the firm's business plans. It should determine the total IT infrastructure costs. If a firm has no IT strategy, it is important to consider developing one which can take into account the firm's five-year strategic plan.

Information technology assessment: A business firm should avoid investing in technology that is not desirable – like those that are still being experimented, or very advanced and expensive above the firm's expenditure cover, or sometimes unreliable. A company should only spend on technologies for which standards have been established and IT vendors are competing on cost, not design, and where there are multiple suppliers.

You may want to note that this is however subjective as sometimes a company may need to spend on new technologies to achieve competitive advantage, develop new business models and capabilities, or improve their system security.

Competitor firm services: It is important for a business firm to assess the technology services that other competitors do offer to their customers, suppliers, and employees. An organisation can establish quantitative and qualitative measures to compare them to those of their firm. If your firm's service levels fall short, your company is at a competitive disadvantage. Look for ways your firm can excel at service levels.

Competitor firm IT infrastructure investments: To be successful, a business firm ought to benchmark its expenditures for IT infrastructure against its competitors. As a firm, you can find such information about your competitors from the information that they publish especially on how they spend on IT investments. Another way is to obtain such information from any public authorized bodies where different companies submit their annual reports.

The Role of Information Systems in Organizations

Strategic Business Objectives of Information Systems

Information systems play a very important role in businesses today. Before a business invests in information systems, it is necessary to determine the main objectives, aims, reasons, or importance as to why they should invest in such systems. There are six (6) primary objectives why companies invest in information systems. They include:

1. Operational Excellence

When a company invests in information systems, it is easy to make higher profits and increase on productivity due to the efficiency that comes with information systems.

2. New Products, Services, and Business Models

A business model describes how a company produces, delivers, and sells products or services to create wealth. Information technology and systems help businesses to come up with new business models, products, or services.

3. Customer and Supplier Intimacy

If customers are served well, it leads to them returning to the same organisation next time they want a similar service or product. This will in the end help the firm to have increased revenue and profits.

In some companies, information systems are used to track customer preferences, and aid in performing more transactions within a short period of time, leading to customer satisfaction and enhanced relationships between firms and suppliers, as well as its customers.

4. Improved Decision Making

Decision making is of paramount importance in every organisation. Managers make decisions from reports that are generated from the information stored in information systems.

Absence of information can lead to managers using guess work to make important decisions. This may lead to issues like misallocation of resources, overproduction or underproduction of goods and services, and sometimes a firm may even lose its customers.

5. Competitive Advantage

Information systems help businesses to have competitive advantage over other companies which do not have them installed. Information systems act as a tool for delivering better performance, charging less for superior products and responding to customers and suppliers in real time.

6. Survival

In the modern world, information technology and systems are very essential in as far as business survival is concerned. Business firms invest in information technologies and systems in order to survive on the market. In other words, information systems are a necessity nowadays if business firms are to survive.

An example can be drawn from a scenario where few banks introduced automated teller machines (ATM) in order to attract more customers, when technology was just booming in Malawi. Other banks followed suit in subsequent years because many customers were rushing to the banks who had ATM machines installed.

It can therefore be concluded that the presence of automated teller machines to offer ATM services to retail customers is simply a requirement for banks to remain and survive in the retail banking business.

Business Drivers of IS/IT

A business driver is a force in the environment to which businesses must respond and that influences the direction of the business. Likewise, examine carefully the inhibitors or negative factors that create management challenges—factors that could scuttle the development of a global business.

A summary of the main business drivers of IS and IT is shown in the table below:

General Cultural Factors	Specific Business Factors
• Global communication and transportation technologies • Development of global culture • Emergence of global social norms • Political stability • Global knowledge base	• Global markets • Global production and operations • Global coordination • Global workforce • Global economies of scale